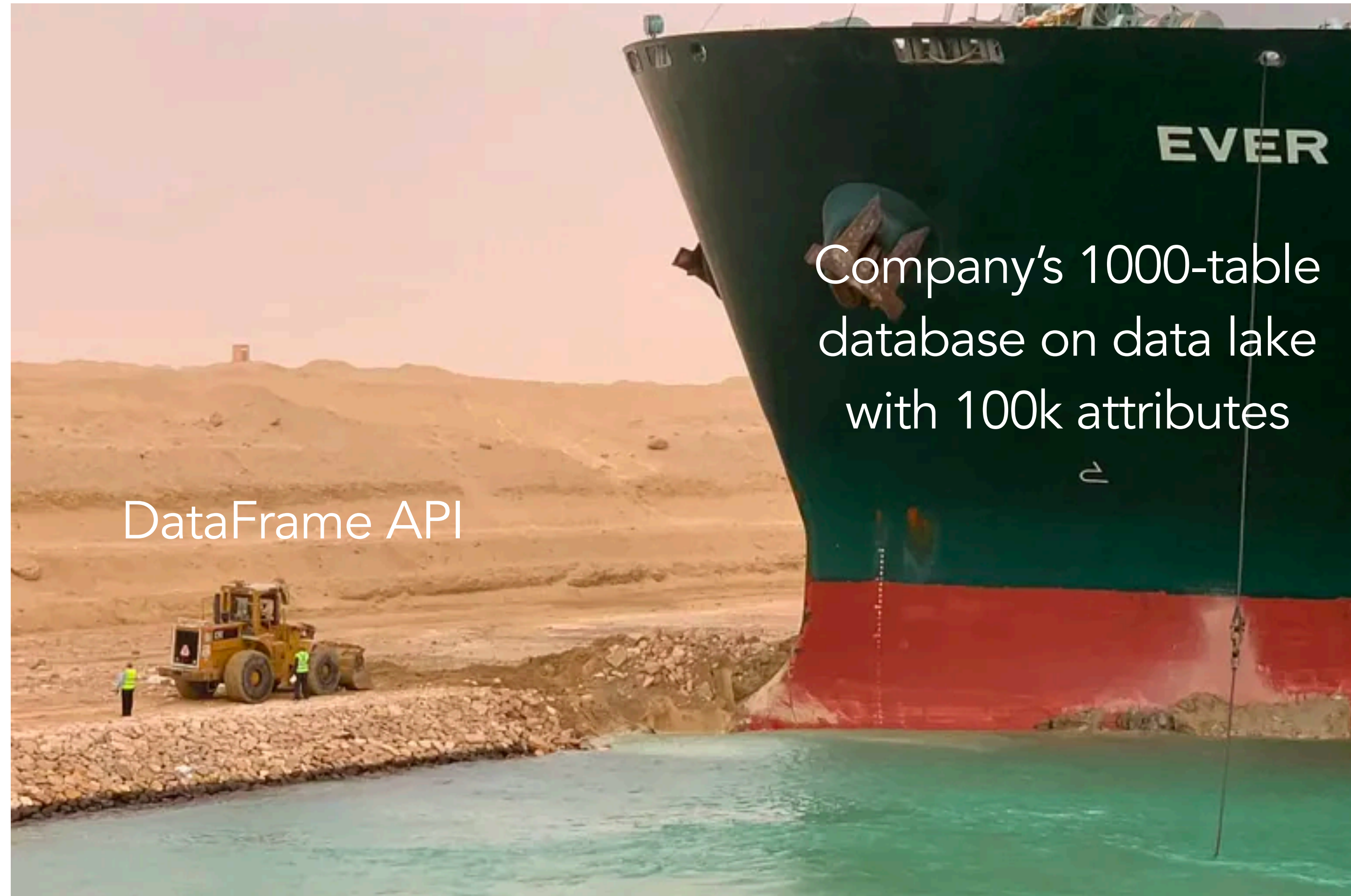


DSC 204a Scalable Data Systems

- Haojian Jin



Company's 1000-table
database on data lake
with 100k attributes

DataFrame API

Question?

Claud Code + DeepSeek v4?

Cheaper?

 DeepSeek API Docs

Quick Start

Your First API Call

Models & Pricing

Token & Token Usage

Rate Limit

Error Codes

API Guides

Thinking Mode

Multi-round Conversation

Chat Prefix Completion (Beta)

FIM Completion (Beta)

JSON Output

Tool Calls

Context Caching

Anthropic API

Agent Integrations

Integrate with Claude Code

Integrate with OpenCode

Integrate with OpenClaw

Contribute Your Agent Integrations

API Reference

News

Other Resources

FAQ

Change Log

English

API Guides

Agent Integrations

Integrate with Claude Code

Integrate with Claude Code

Claude Code is an AI coding assistant that runs in the terminal.

1. Install Claude Code

- Install [Node.js](#) 18+.
- Windows users need to install [Git for Windows](#).
- Run the following command in your terminal to install Claude Code:

```
npm install -g @anthropic-ai/claude-code
```

- After installation, run the following command. If the version number is displayed, the installation is successful:

```
claude --version
```

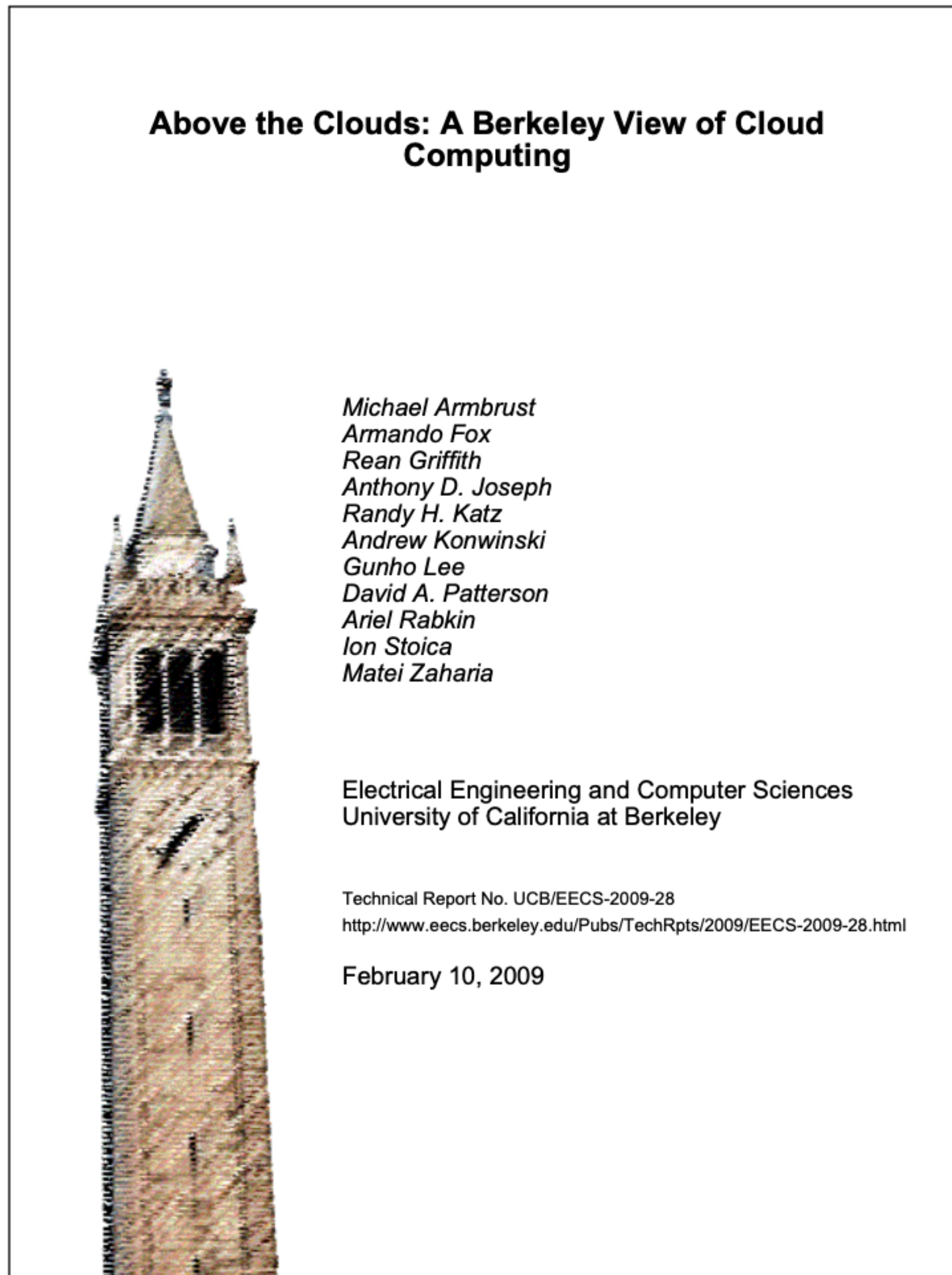
2. Configure Environment Variables

Linux / Mac users, run the following commands to configure environment variables for the [DeepSeek Anthropic API](#). Get your API Key from the [DeepSeek Platform](#):

```
export ANTHROPIC_BASE_URL=https://api.deepseek.com/anthropic
export ANTHROPIC_AUTH_TOKEN=<your DeepSeek API Key>
export ANTHROPIC_MODEL=deepseek-v4-pro[1m]
export ANTHROPIC_DEFAULT_OPUS_MODEL=deepseek-v4-pro[1m]
export ANTHROPIC_DEFAULT_SONNET_MODEL=deepseek-v4-pro[1m]
export ANTHROPIC_DEFAULT_HAIKU_MODEL=deepseek-v4-flash
export CLAUDE_CODE_SUBAGENT_MODEL=deepseek-v4-flash
export CLAUDE_CODE_EFFORT_LEVEL=max
```

Windows users, run:

Suggested reading



 University of California, Berkeley
<https://www2.eecs.berkeley.edu> › EECS-2009-28

Above the Clouds: A Berkeley View of Cloud Computing
by M Armbrust · 2009 · Cited by 9289 — Our goal in this paper to clarify terms, provide simple formulas to quantify comparisons between of cloud and conventional Computing, and ...
25 pages

Today's topic

- Cloud computing evolution - sharing granularity
- Cloud computing layers
- Advantages of Cloud computing

Cloud Computing

- Compute, storage, memory, networking, etc. are **virtualized** and exist on **remote** servers; **rented** by application users
- The opposite:
 - On-premises refers to IT infrastructure hardware and software applications that are hosted on-site.

Cloud Computing v.s. on-premise clusters

- Compute, storage, memory, networking, etc. are virtualized and exist on *remote servers*; *rented* by application users
- Main pros of cloud vs on-premise clusters:
 - **Manageability**: Managing hardware is not user's problem
 - **Pay-as-you-go**: Fine-grained pricing economics based on actual usage (granularity: seconds to years!)
 - **Elasticity**: Can dynamically add or reduce capacity based on actual workload's demand
- Infrastructure-as-a-Service (IaaS); Platform-as-a-Service (PaaS); Software-as-a-Service (SaaS)

Evolution of Cloud Infrastructure

- **Data Center:** Physical space from which a cloud is operated
- **3 generations of data centers/clouds:**
 - Cloud 1.0 (Past)
 - Cloud 2.0 (Current)
 - Cloud 3.0 (Ongoing Research)

Car Analogy



Own a car
(Bare metal servers)



Rent a car
(VPS)



City car-sharing
(Serverless)

Cars are parked **95%** of the time (loige.link/car-parked-95)

How much do you use the car?

Cloud 1.0 (Past)

- Networked servers;
- User rents servers (time-sliced access) needed for data/software

From Lecture 5:

Virtualization of Hardware Resources

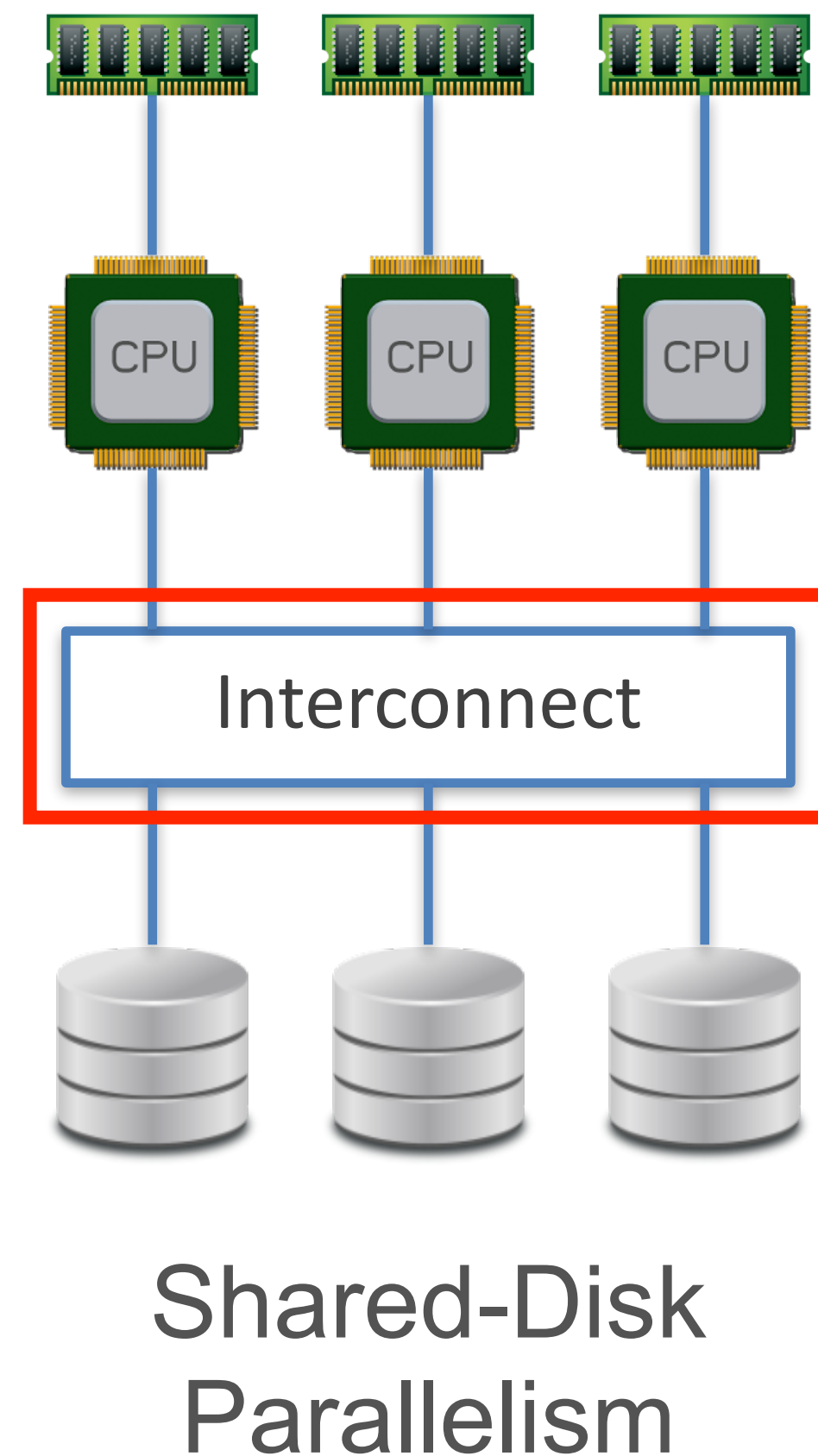
Q: But is it not risky/foolish for OS to hand off control of hardware to a process (random user-written program)?!

- OS has *mechanisms* and *policies* to regain control
- **Virtualization:**
 - Each hardware resource is treated as a virtual entity that OS can divvy up among processes in a controlled way
- **Limited Direct Execution:**
 - OS mechanism to time-share CPU and preempt a process to run a different one, aka “context switch”
 - A **Scheduling policy** tells OS what time-sharing to use
 - Processes also must transfer control to OS for “privileged” operations (e.g., I/O); **System Calls API**

Cloud 2.0 (Current)

- “Virtualization” of networked servers;
- User rents amount of resource capacity (e.g., memory, disk);
- Cloud provider has a lot more flexibility on provisioning (multi-tenancy, load balancing, more elasticity, etc.)

Parallelism in the Cloud

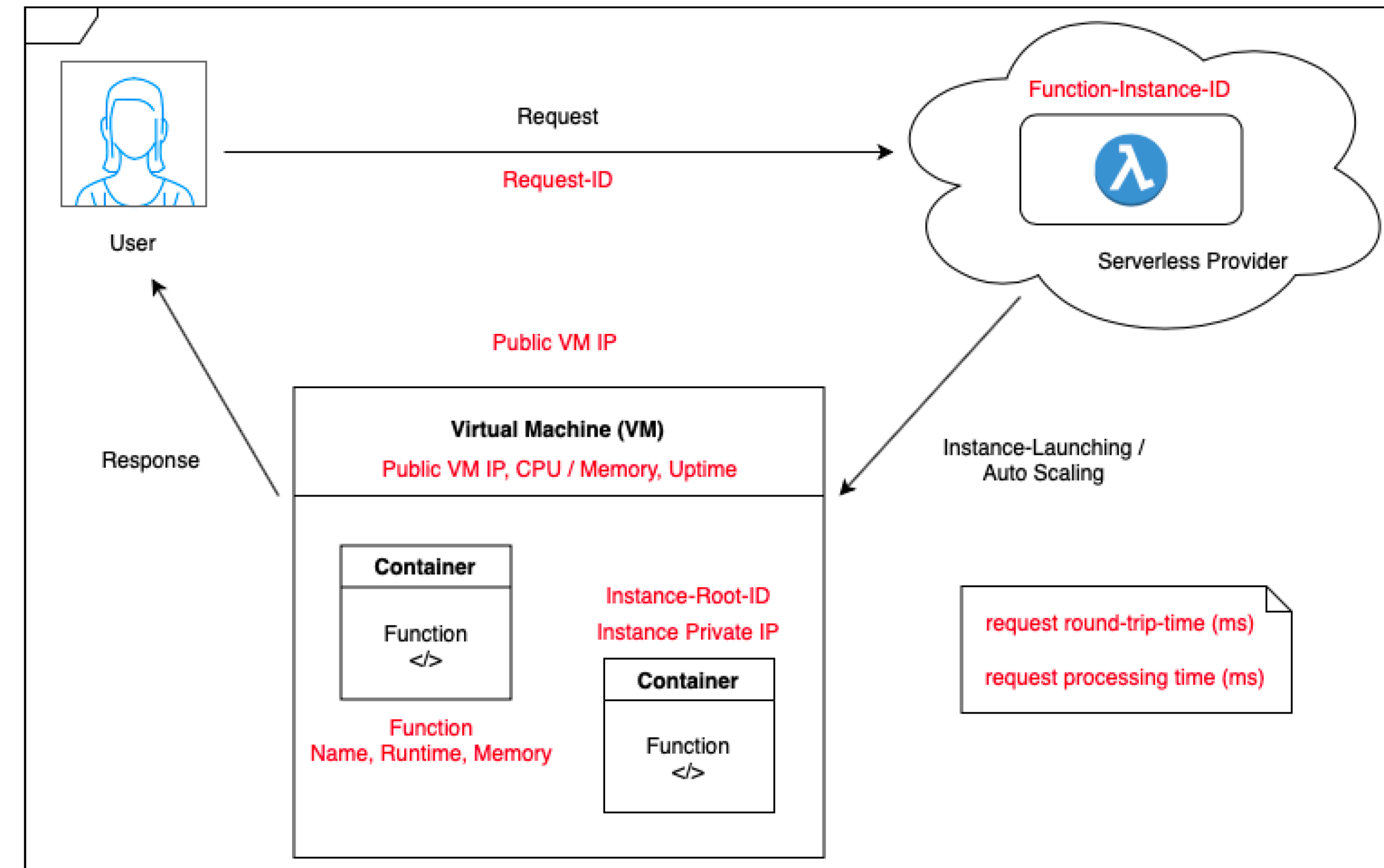


Modern networks in data centers have become much faster: 100GbE to even TbE!

- **Decoupling** of compute+memory from storage is common in cloud
 - *Hybrids* of shared-disk parallelism + shared-nothing parallelism
 - E.g, store datasets on S3 and read as needed to local EBS

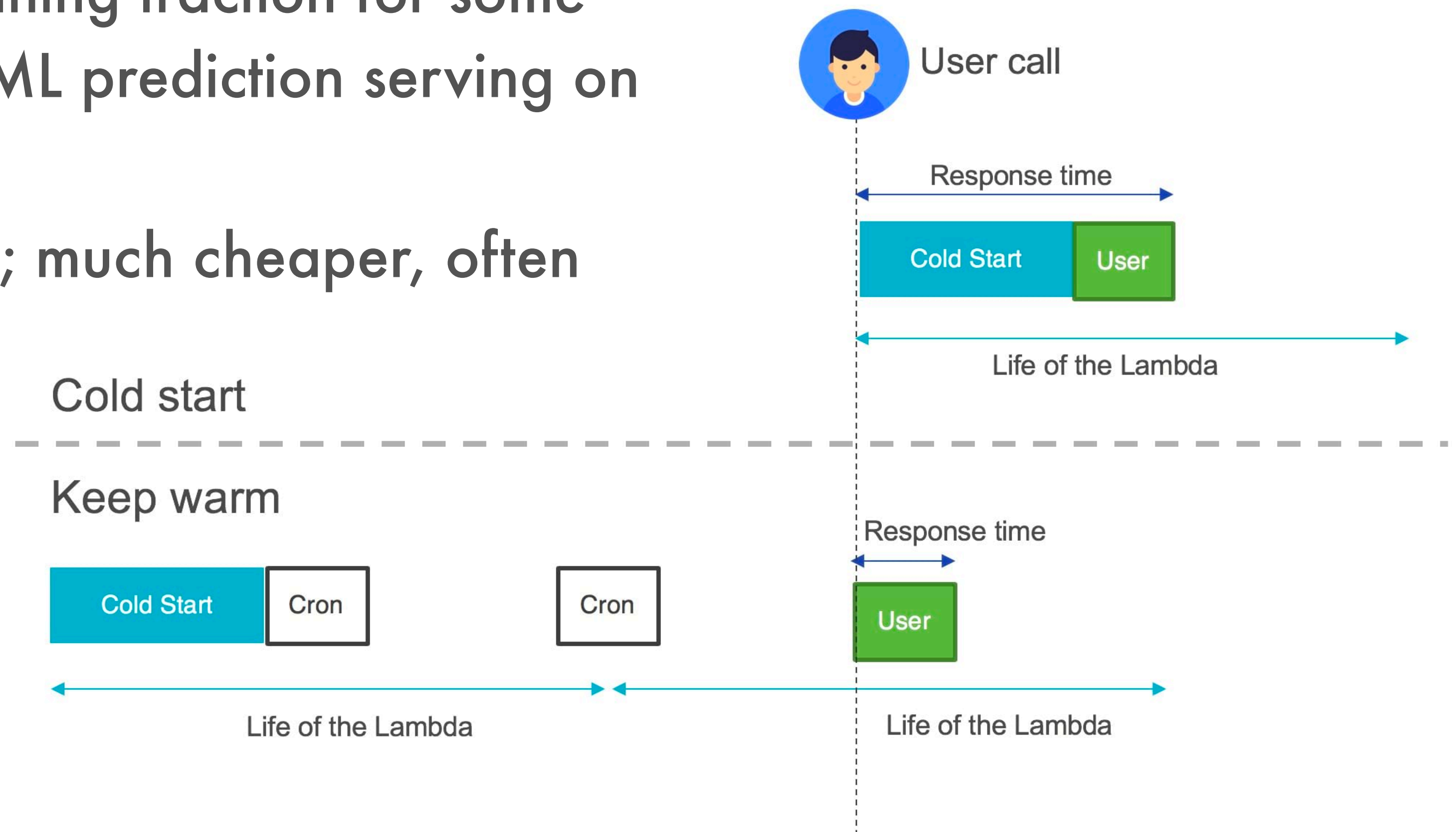
Cloud 3.0 (Ongoing Research)

- Full **resource disaggregation**! That is, compute, memory, storage, etc. are all network-attached and elastically added/removed
- User gives a program (function) to run and specifies CPU and DRAM needed
- Cloud provider abstracts away all resource provisioning entirely
- Aka *Function-as-a-Service* (FaaS)



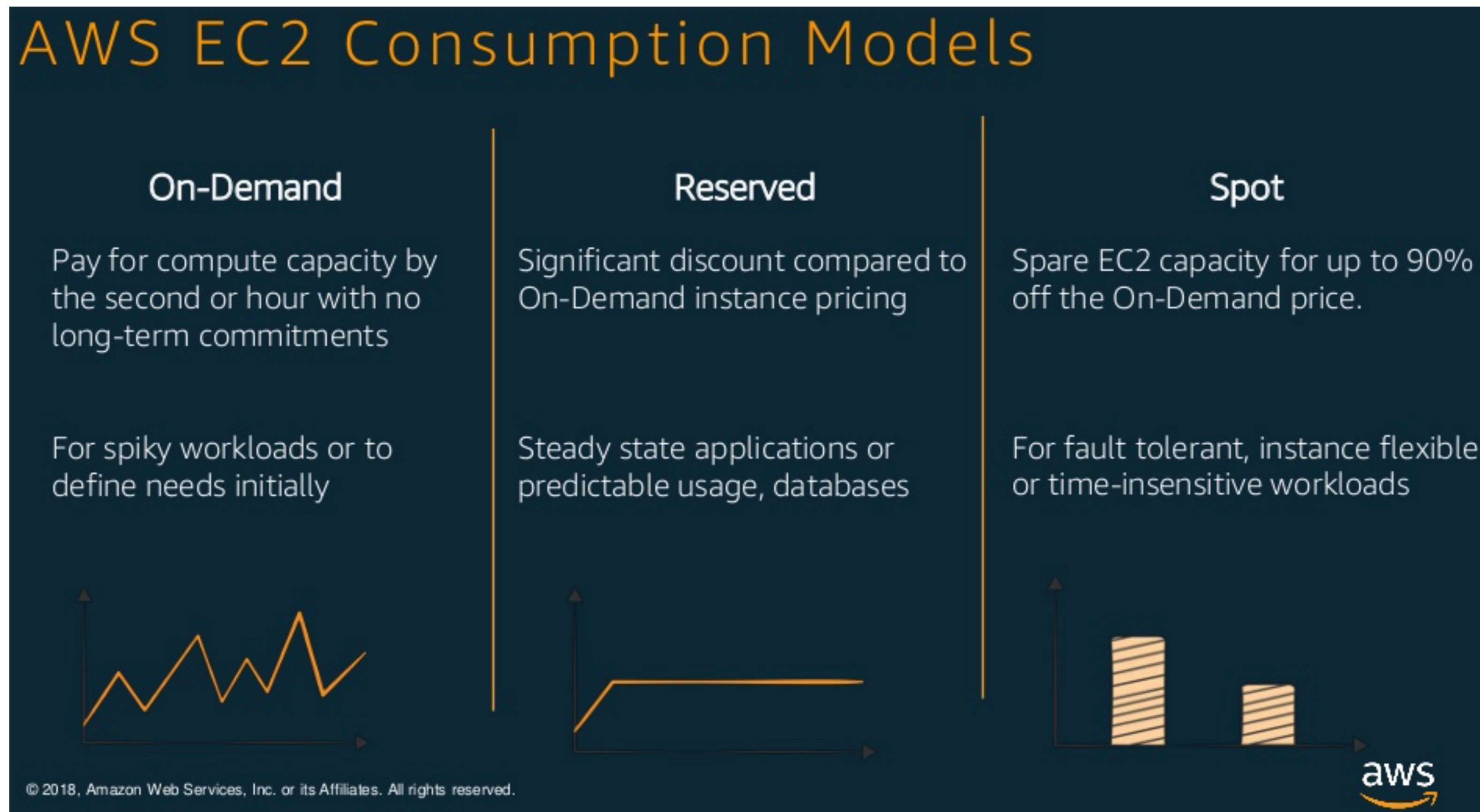
Cloud 3.0 (Ongoing Research)

- “Serverless” and disaggregated resources all connected to fast networks
- **Serverless** paradigm gaining traction for some applications, e.g., online ML prediction serving on websites
- Higher resource efficiency; much cheaper, often by 10x vs Spot instances



New Cloud Renting Paradigms

- ❖ Cloud 2.0's flexibility enables radically different paradigms
- ❖ AWS example below; Azure and GCP have similar gradations



More on Spot vs On-Demand

	Spot Instances	On-Demand Instances
Launch time	Can only be launched immediately if the Spot Request is active and capacity is available.	Can only be launched immediately if you make a manual launch request and capacity is available.
Available capacity	If capacity is not available, the Spot Request continues to automatically make the launch request until capacity becomes available.	If capacity is not available when you make a launch request, you get an insufficient capacity error (ICE).
Hourly price	The hourly price for Spot Instances varies based on demand.	The hourly price for On-Demand Instances is static.
Rebalance recommendation	The signal that Amazon EC2 emits for a running Spot Instance when the instance is at an elevated risk of interruption.	You determine when an On-Demand Instance is interrupted (stopped, hibernated, or terminated).
Instance interruption	You can stop and start an Amazon EBS-backed Spot Instance. In addition, the Amazon EC2 Spot service can interrupt an individual Spot Instance if capacity is no longer available, the Spot price exceeds your maximum price, or demand for Spot Instances increases.	You determine when an On-Demand Instance is interrupted (stopped, hibernated, or terminated).

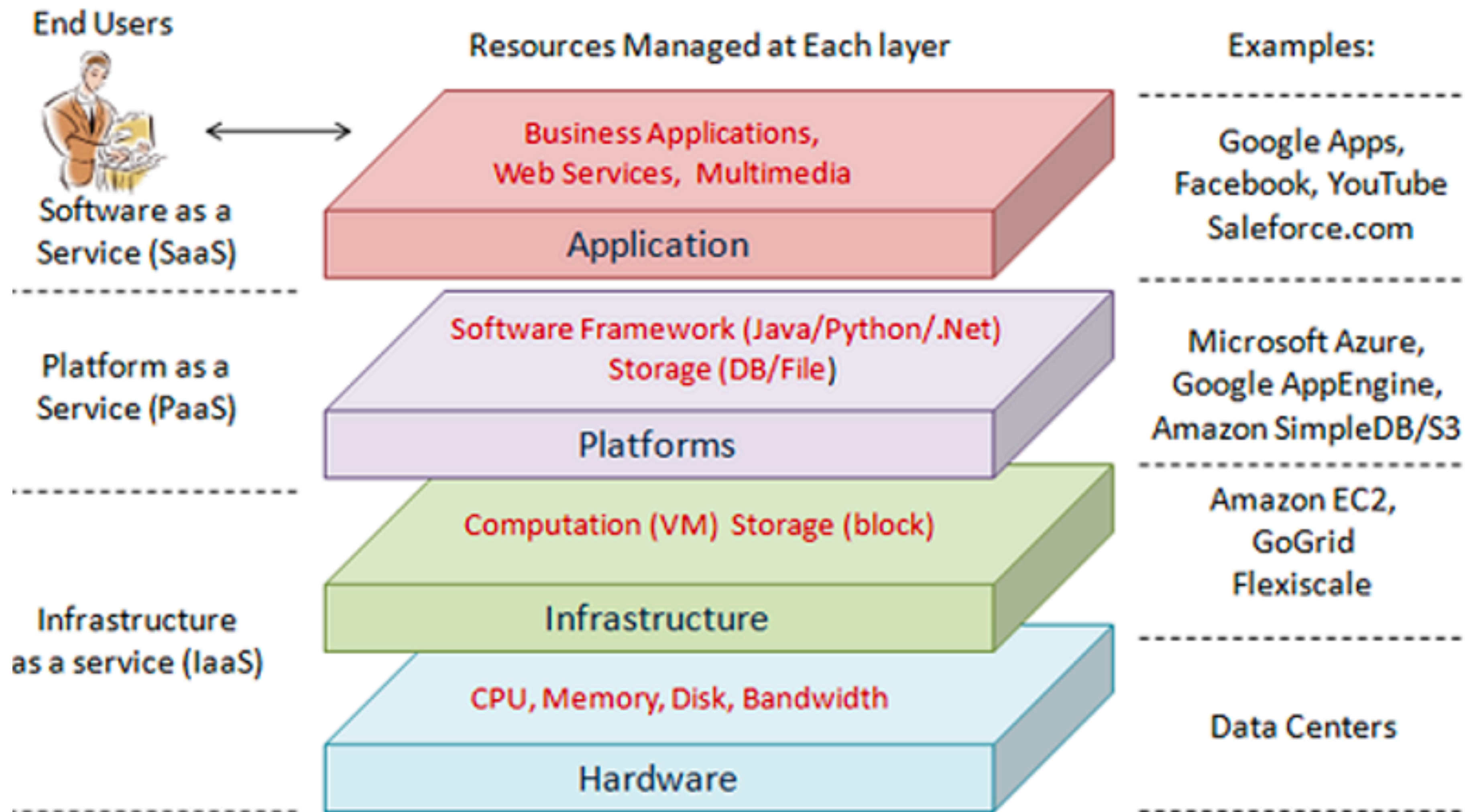
Advantage and disadvantage

- Cloud 1.0:
 - +: Simple, Perfect isolation,
 - - : Expensive.
- Cloud 2.0:
 - +: Cheaper than Cloud 1.0.
 - - : Some resource waste
- Cloud 3.0:
 - +: Cheapest
 - - : Cold-start issues, Security & Privacy, Hard to manage.

Today's topic

- Cloud computing evolution - sharing granularity
- Cloud computing layers
- Cloud computing practices

Cloud Computing



Featured Services	Analytics		Resources and Media	
	Amazon Athena		Blog	
	Query data in S3 using SQL		Read the latest blog posts on Big Data	
	Amazon CloudSearch		What's New on AWS	
	Managed search service		See announcements for Big Data and Analytics on AWS	
	Amazon DataZone (Preview)		Customer Enablement	
	Unlock data across organizational boundaries with built-in governance		AWS IQ	
	Amazon OpenSearch Service		Complete your projects faster with help from AWS Certified third-party experts	
	Search, visualize, and analyze up to petabytes of text and unstructured data		AWS Managed Services	
	Amazon EMR		Operate your AWS infrastructure for you	
	Easily run big data frameworks		AWS Professional Services	
	Amazon FinSpace		Accelerate your business outcomes with AWS	
	Analytics for the financial services industry		AWS Training and Certification	
	Amazon Kinesis		Build and validate your skills and technical expertise	
	Analyze real-time video and data streams			
	Amazon Managed Streaming for Apache Kafka			
	Fully managed Apache Kafka service			
	Amazon Redshift			
	Fast, simple, cost-effective data warehousing			
	Amazon QuickSight			
	Fast business analytics service			
	AWS Clean Rooms			
	Match, analyze, and collaborate on datasets—without sharing or revealing underlying data			
	AWS Data Exchange			
	Find, subscribe to, and use third-party data in the cloud			
	AWS Data Pipeline			
	Orchestration service for periodic, data-driven workflows			
	AWS Glue			
	Simple, scalable, and serverless data integration			
	AWS Lake Formation			
	Build, manage, and secure your data lake			

Examples of AWS Cloud Services

- **IaaS:**
 - **Compute:** EC2, ECS, Fargate, Lambda
 - **Storage:** S3, EBS, EFS, Glacier
 - **Networking:** CloudFront, VPC
- **PaaS:**
 - **Database/Analytics Systems:** Aurora, Redshift, Neptune, ElastiCache, DynamoDB, Timestream, EMR, Athena
 - **Blockchain:** QLDB; **IoT:** Greengrass
- **SaaS:**
 - **ML/AI:** SageMaker, Elastic Inference, Lex, Polly, Translate, Transcribe, Textract, Rekognition, Ground Truth
 - **Business Apps:** Chime, WorkDocs, WorkMail

Today's topic

- Cloud computing evolution - sharing granularity
- Cloud computing layers
- **Advantages of Cloud computing**

OMG, is all this complexity worth it?!

- Depends on user's/application's Pareto tradeoffs! :)
- **On-premise** cluster are still common in large enterprises, healthcare, and academia; “hybrid clouds” too
- Recall main pros of cloud: manageability, cost, and elasticity
- Some main cons of cloud (vs on-premise):
 - **Complexity** of composing cloud APIs and licenses; data scientists must keep relearning; “CloudOps” teams
 - **Cost** over time can *crossover* and make it costlier!
 - Easier to **waste money** accidentally on the fly
 - “**Lock-in**” by cloud vendor
 - **Privacy, security, and governance** concerns
 - **Internet disruption or unplanned downtime**, e.g., AWS outage in 2015 made Netflix, Tinder, etc. unavailable! :)

Recap: Cloud Computing v.s. on-premise clusters

- Compute, storage, memory, networking, etc. are virtualized and exist on *remote servers*; *rented* by application users
- Main pros of cloud vs on-premise clusters:
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Review Questions

1. What are the 3 main layers of a typical cloud? Give examples of AWS services in each layer. Which ones do your PAs use?
2. What is a benefit of separating PaaS from SaaS in cloud?
3. Briefly explain 1 pro and 1 con of On-Demand vs Spot instances on AWS.
4. What is so great about the serverless cloud anyway?
5. Briefly explain 2 pros and 2 cons of cloud vs on-premise clusters.